DSA0603 Data Handling and Visualization

Slot B

CO2 – Assessment Test 4 (AT1)

Chart Construction Test

Name : B Karunyeswari

Reg No : 192424235

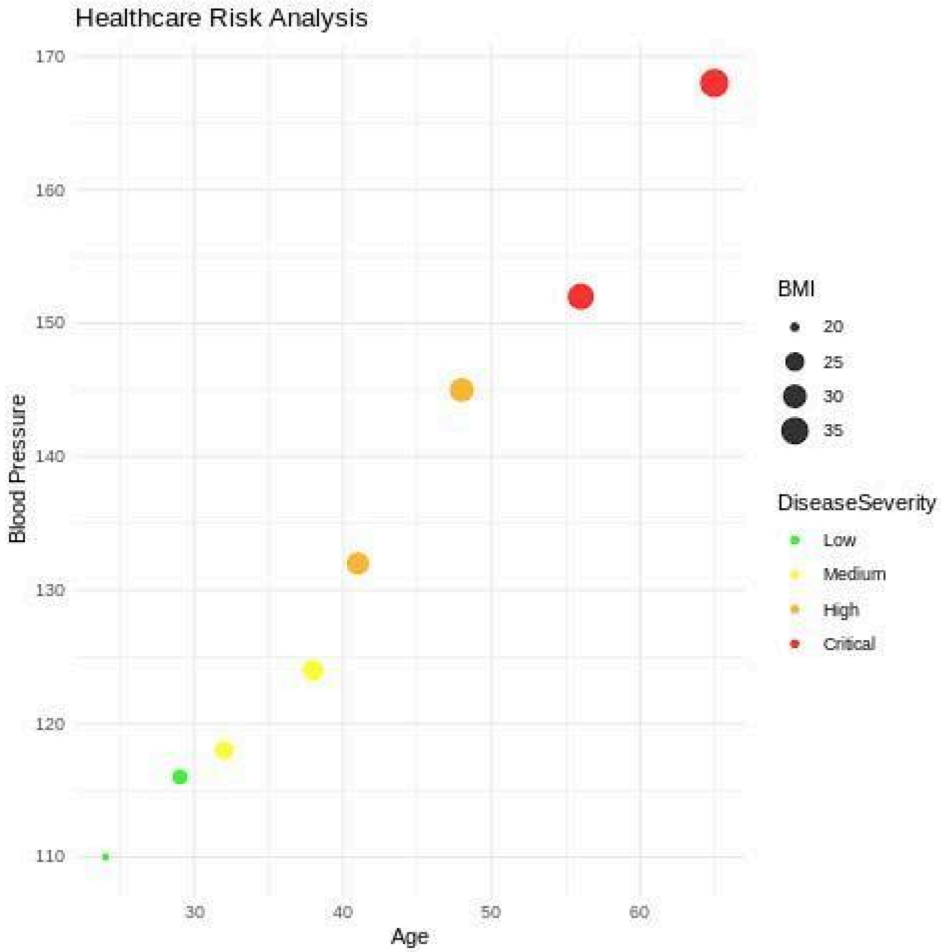
Question 1 (Healthcare Analytics):

R Code:

```
library(ggplot2) health
<- data.frame(
  Age = c(24,32,41,48,56,65,38,29),
  BMI = c(19.8,24.5,28.1,30.3,33.2,36.5,26.2,22.1),
  BloodPressure = c(110,118,132,145,152,168,124,116),
  DiseaseSeverity = factor(c("Low","Medium","High","High",
                             "Critical","Critical","Medium","Low"),
levels=c("Low","Medium","High","Critical"))
)
ggplot(health,
aes(x=Age,
y=BloodPressure,
color=DiseaseSeverity,
size=BMI))+
  geom_point(alpha=0.8)+

scale_color_manual(values=c("green","yellow","orange","red"+
labs(title="Healthcare Risk Analysis",        x="Age",
y="Blood Pressure")+   theme_minimal()
```

Output:



Justification

- Age → X-axis: Independent health factor.
- Blood Pressure → Y-axis: Indicates cardiovascular risk.
- Disease Severity → Color: Green to Red clearly shows increasing health risk.
- BMI → Point Size: Larger circles represent higher BMI, making obesity easy to identify.
- The visualization helps doctors quickly detect high-risk patients.

Question 2 (Climate Science):

R Code: Dataset

library(ggplot2)

climate <-

data.frame(

Month=factor(month.

```
abb,
levels=month.abb),
SolarRadiation=c(180,200,240,280,320,300,260,250,240,220,190,170),
Temperature=c(20,23,29,35,39,36,32,31,30,28,24,21),
Rainfall=c(25,18,12,8,5,65,150,170,120,85,45,30)
)
```

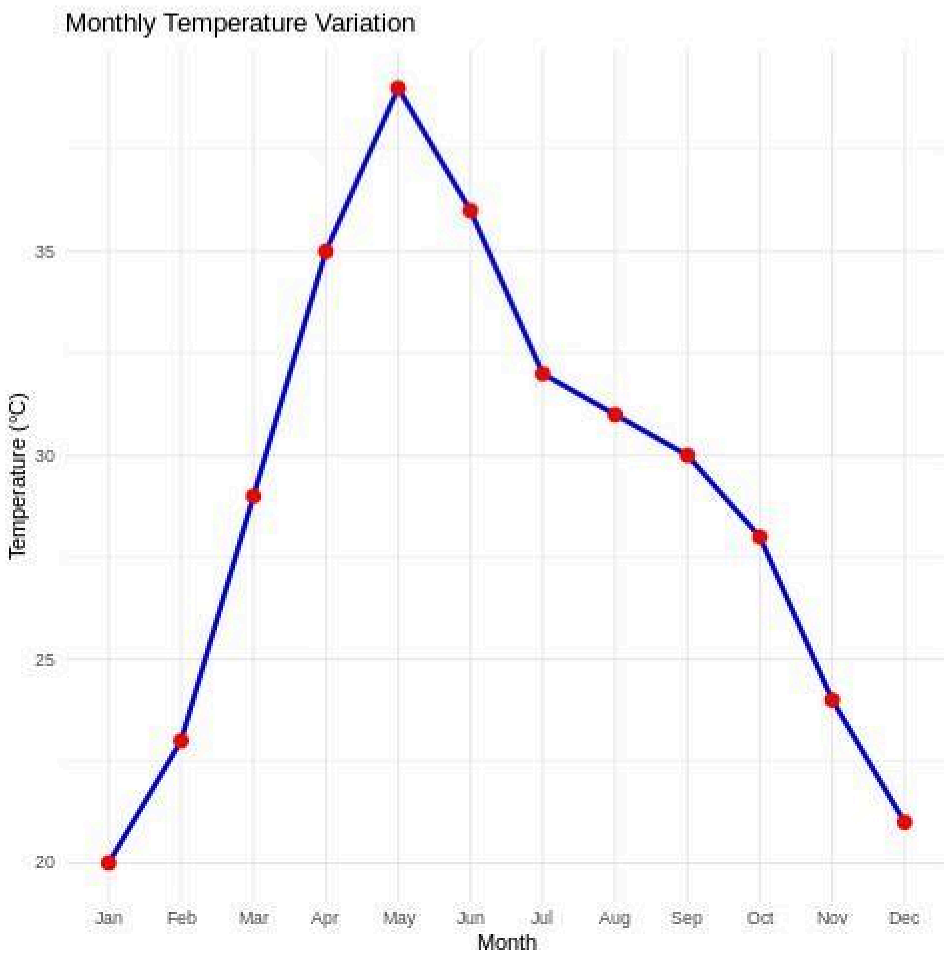
1. Cartesian Line Chart

```
ggplot(climate,
aes(x=Month,y=Temperature,group=1))+
geom_line(color="blue",size=1.2)+
geom_point(color="red",size=3)+
labs(title="Monthly Temperature
Variation", x="Month", y="Temperature
(°C)")+ theme_minimal()
```

2. Polar Coordinate Chart

```
ggplot(climate, aes(x=Month, y=Rainfall,
fill=Rainfall))+
geom_bar(stat="identity")+ coord_polar()+
scale_fill_gradient(low="lightblue",
high="darkblue")+ labs(title="Annual
Rainfall Distribution")+ theme_minimal()
```

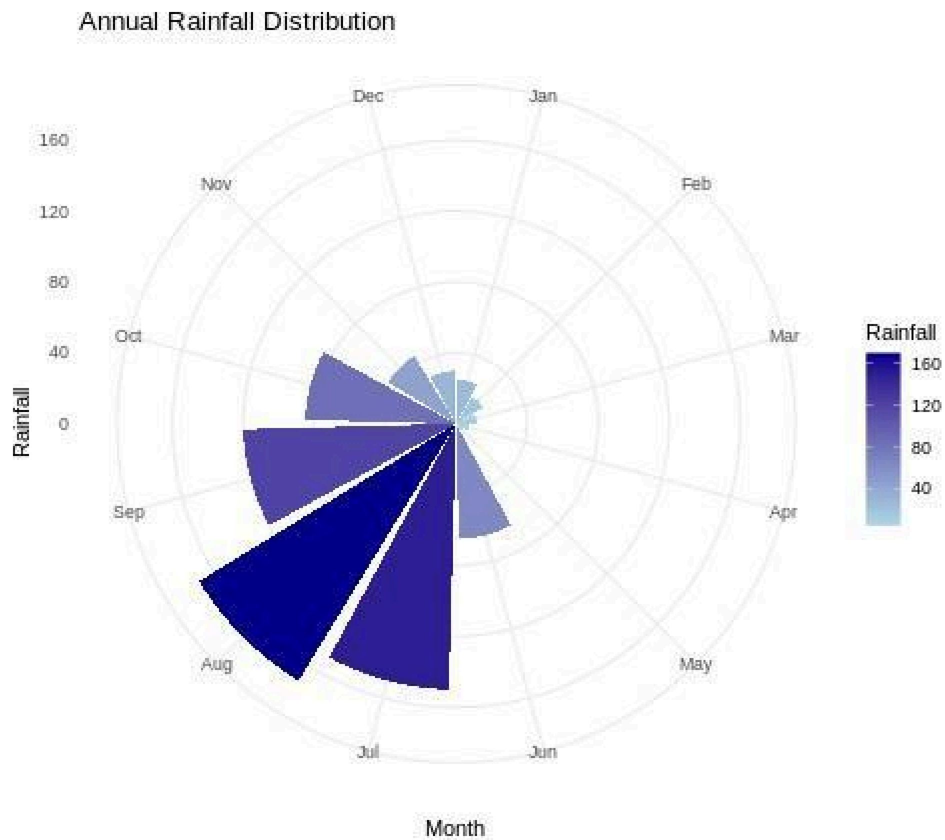
OUTPUT:



2. Polar Coordinate Chart

```
ggplot(climate, aes(x=Month, y=Rainfall,
fill=Rainfall))+
geom_bar(stat="identity")+
coord_polar()+
scale_fill_gradient(low="lightblue",
high="darkblue")+ labs(title="Annual
Rainfall Distribution")+ theme_minimal()
```

OUTPUT:



Comparison:

Cartesian Chart:

Advantages

- Easy to observe trends.
- Precise value comparison.
- Suitable for continuous variables.

Limitations

- Does not emphasize cyclic nature.

Polar Chart:

Advantages:

- Clearly represents seasonal cycles.

- Attractive visualization for yearly rainfall.

Limitations:

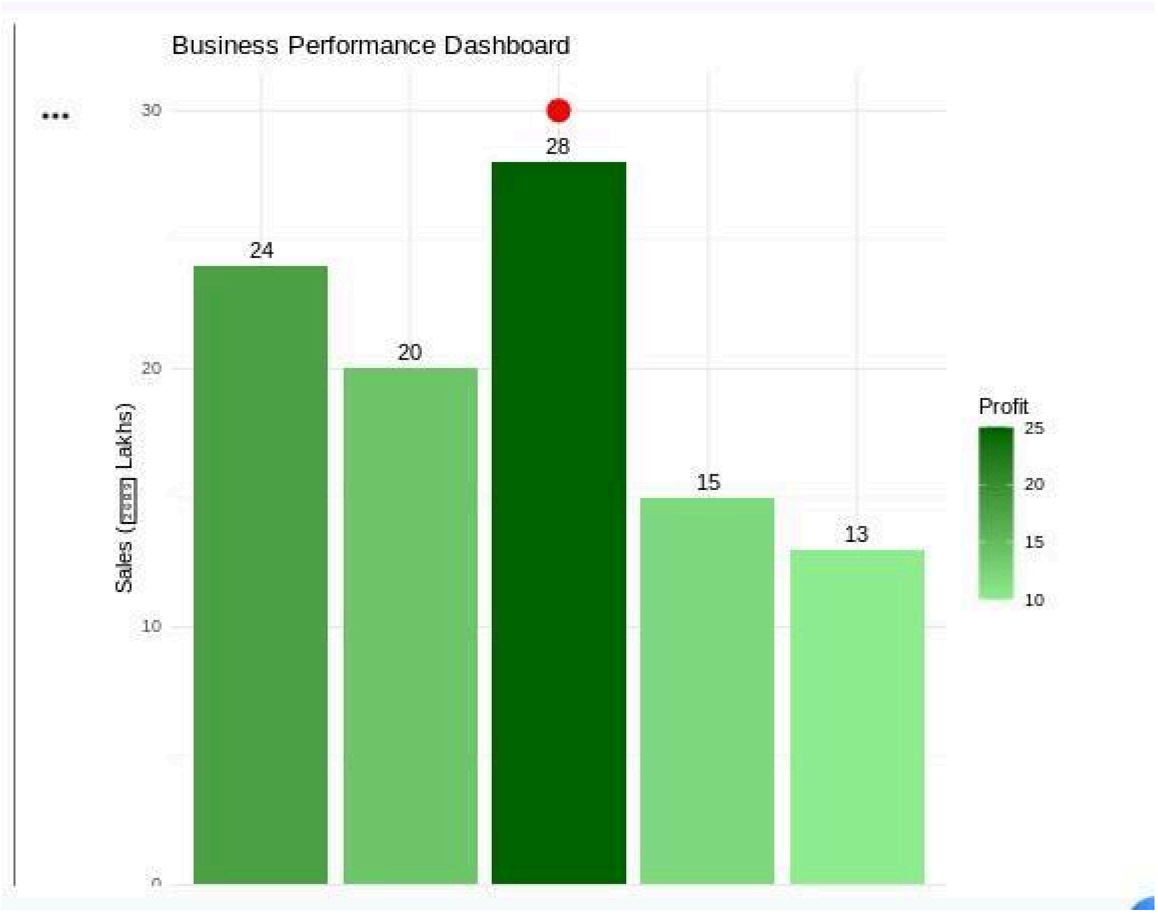
- Harder to compare exact values.
- Small differences may appear distorted.

Question 3 (Business Analytics & Marketing):

R Code :

```
library(ggplot2)
business <- data.frame(
Product=c("A","B","C","D","E"),
Sales=c(24,20,28,15,13),
Profit=c(18,14,25,12,10),
MarketShare=c(210,185,260,145,120)
)
business$Highlight <- ifelse(
business$MarketShare==max(business$MarketShare),
"Highest","Others")
ggplot(business,
aes(x=Product,
y=Sales,
fill=Profit))+
geom_bar(stat="identity")+
geom_text(aes(label=Sales),vjust=-0.5)+
scale_fill_gradient(low="lightgreen", high="darkgreen")+
geom_point(data=subset(business,Highlight=="Highest")
, aes(x=Product,y=Sales+2), color="red",size=5)+
labs(title="Business Performance Dashboard", y="Sales
(₹ Lakhs)")+ theme_minimal()
```

OUTPUT:



Justification

- Bar Height: Represents sales volume clearly.
- Sequential Green Palette: Darker shades indicate higher profit.
- Red Highlight: Draws attention to the highest market share product.
- Helps managers identify both profitability and market leadership quickly.

Question 4 (Transportation Engineering): R

Code

```
library(ggplot2) traffic

<- data.frame(

Direction=c("North","North-East","East",

"South-East","South",

"South-West","West",
```

"North-West"),

Vehicles=c(980,1240,1420,860,

760,890,1130,720)

) ggplot(traffic, aes(x=Direction,

y=Vehicles, fill=Vehicles))+

geom_bar(stat="identity")+

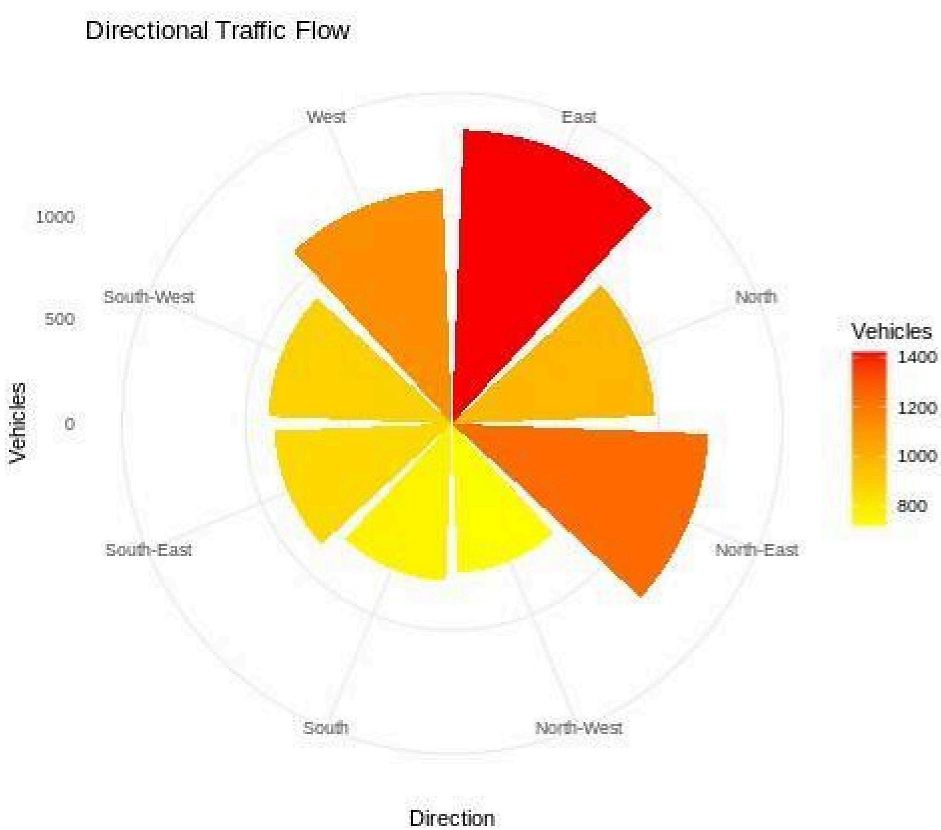
coord_polar()+

scale_fill_gradient(low="yellow",

high="red")+ labs(title="Directional

Traffic Flow")+ theme_minimal()

Output:



Justification

- Polar charts naturally represent compass directions.
- Colors distinguish traffic density.
- High-density roads appear in darker colors.
- Better than Cartesian charts because directional data is circular rather than linear.

Question 5(Environmental Sustainability Dashboard):

Dataset library(ggplot2) library(gridExtra) env <-

data.frame(

City=c("A","B","C","D","E","F"),

AQI=c(48,82,135,165,72,95),
GreenCover=c(45,32,21,18,38,30),

CO2=c(1.8,3.5,5.6,7.4,2.9,4.2),

Population=c(9,15,22,28,12,18)

)

env$AQICategory <- cut(env$AQI,

breaks=c(0,50,100,150,200),

labels=c("Good","Moderate",

"Unhealthy",

"Very Unhealthy")

)

Visualization 1 p1 <-

ggplot(env, aes(x=Population,

y=AQI, color=AQICategory,

size=GreenCover))+

geom_point()+

scale_color_manual(values=c

(

"green",

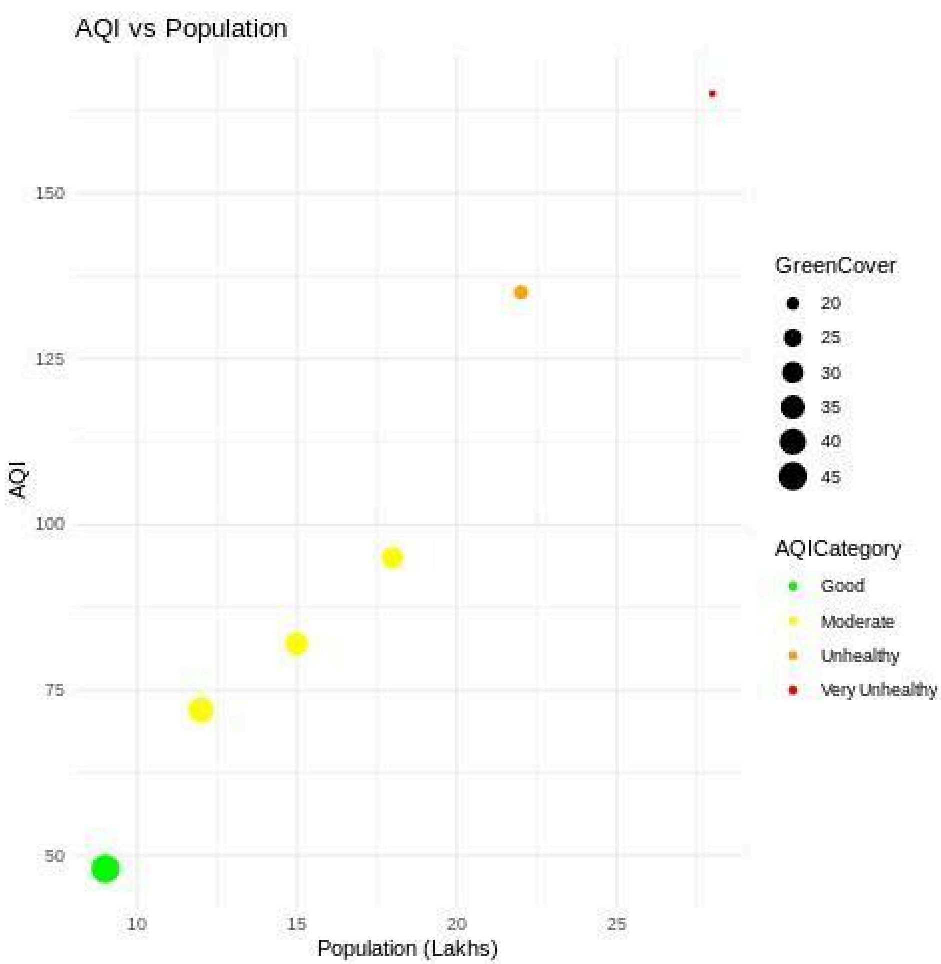
"yellow",

"orange", "red"))+

labs(title="AQI vs

Population")

Output:



Visualization 2 : p2 <- ggplot(env,

aes(x=City, y=CO2, fill=CO2))+

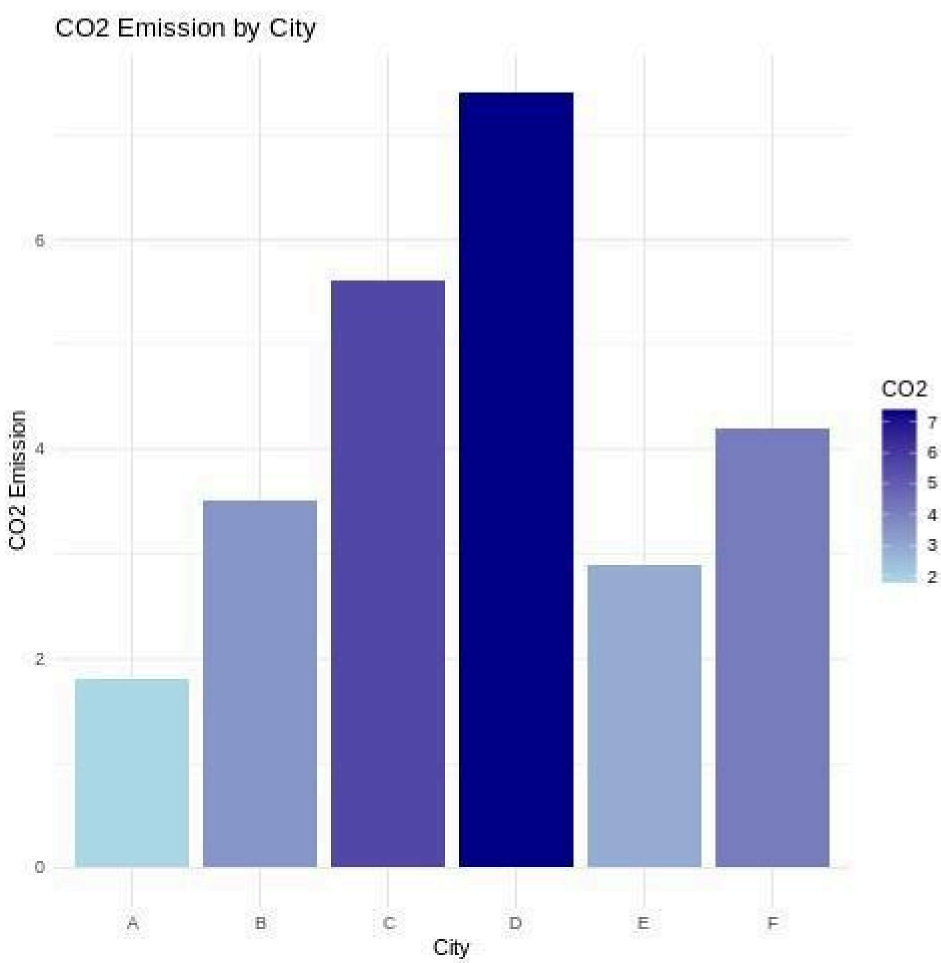
geom_bar(stat="identity")+

scale_fill_gradient(low="lightblue"

, high="darkblue")+

labs(title="CO_2 Emission by City")

Output:



Design Justification

- Population → X-axis: Represents city size.
- AQI → Y-axis: Measures air quality.
- Green Cover → Point Size: Larger circles indicate more vegetation.
- AQI Category → Color: Green, Yellow, Orange, and Red follow standard air quality categories, making pollution levels easy to interpret.
- CO_2 Bar Chart: Enables quick comparison of emissions across cities.

- Blue Gradient: Darker shades indicate higher CO_2 emissions.
- Dashboard Layout: Displays pollution, population, green cover, and emissions together for a comprehensive sustainability assessment.